

K-means and hierarchical clustering [MCQ] (Version : 0)

TEST

 **Correct Answer**

 Answered in 3.35 Minutes

Question 1/20

Before we dive in, it's crucial to acquaint ourselves with the dataset through exploration and preliminary analysis by understanding its characteristics.

Print the descriptive statistics of our dataset.

True or false? The `Spending_Score` is a number from 1 - 50.

☐ True

☒ False

Explanation:

True: This option is incorrect because, according to the results of `data.describe()`, the minimum `Spending_Score` is 1 and the maximum spending score is 99; therefore, it is a number from at least 1 to 99.

Question 2/20

What is the total number of observations in this dataset?

☐ 150

☒ 200

☐ 100

☐ 250

Explanation:

100, 150, 250: These options are incorrect because the descriptive statistics provided in the describe() method output indicate 200 observations in the dataset, as shown by the "count" row for each variable.

Question 3/20

Next, we need to check if the data contains any null values because missing values can disrupt the modelling process by causing errors or biases in our analysis.

Write code that will check if our dataset contains any null values. If so, fill them with zero's.

True or false? Our dataset contains no null values.

☒ True

☐ False

Explanation:

False: This option is incorrect because according to the results of `data.isnull().sum()`, all of our columns contain zero null values.

Question 4/20

Since we are interested in segmenting our customers based on their annual income and spending score, let's visualise the relationship between this pair.

Generate a scatter plot showing the relationship between annual income and spending score for each customer in the dataset.

What insights can be inferred from the scatter plot?

☐ The data points exhibit a uniform distribution across the grid.

☐ There is no relationship between annual income and spending score.



There are visible clusters forming, particularly in the center of the plot.



Points are evenly spread, indicating no discernible patterns or correlations.

Explanation:

"The data points exhibit a uniform distribution across the grid." This option is incorrect because a uniform distribution across the grid implies an equal density of data points in all areas of the plot, which contradicts the plot which has visible clusters, indicating non-uniform distribution.

"Points are evenly spread, indicating no discernible patterns or correlations." This option is incorrect because evenly spread points suggest no clustering or discernible patterns, which contradicts the plot.

"There is no relationship between annual income and spending score" This option is incorrect because the presence of clusters indicates that there is a relationship or some level of association between annual income and spending score among the data points.

Question 5/20

Before continuing our analysis, we need to scale our data to prevent attributes with larger magnitudes from dominating the distance calculations.

The code attempts to scale the data but it returns an error. Which of the following best explains why the provided code returns an error?



The code fails because the MinMaxScaler requires the input data to be a NumPy array, and the provided data is not in this format.



The code attempts to fit the MinMaxScaler on non-numeric data, resulting in a ValueError because the MinMaxScaler only works with numerical data.



The error occurs because the MinMaxScaler cannot handle missing

☐ values in the dataset.

☐ The code fails because the MinMaxScaler requires the data to be in a specific range, and the provided data exceeds this range.

Explanation:

"The code fails because the MinMaxScaler requires the data to be in a specific range, and the provided data exceeds this range." This option is incorrect because the provided error message does not suggest that the data exceeds the range expected by the MinMaxScaler. The error mentions a problem converting a string to a float, indicating non-numeric data.

"The error occurs because the MinMaxScaler cannot handle missing values in the dataset." This option is incorrect because the error message does not indicate any issues related to missing values in the dataset. The error is related to converting strings to floats, suggesting that the problem lies with non-numeric data rather than missing values.

"The code fails because the MinMaxScaler requires the input data to be a NumPy array, and the provided data is not in this format". This option is incorrect because the MinMaxScaler from Sklearn is designed to work with various data structures, including dataframes.

Question 6/20

Let's initiate our cluster analysis by selecting the features for segmentation. Given our focus on customer segmentation using their annual income and spending score, we'll designate the `Annual_Income_(k$)` and `Spending_Score` columns as the features for clustering, assigning them to the variable `X`.

Following feature selection, let's scale our data using the MinMaxScaler as attempted in the previous question and implement the K-Means clustering algorithm on the scaled dataset, opting for 5 clusters based on an initial analysis of our scatter plot.

Following the application of the K-Means model using 5 clusters, which of the following code snippets will display the coordinates of the centroids representing the clusters identified by the algorithm?

☒ `print(kmeans.cluster_centers_)`

☐ `print(kmeans.centroids)`

☐ `print(kmeans.cluster_centers)`

☐ `print(kmeans.centroids_)`

Explanation:

`print(kmeans.centroids_)` - This option is incorrect because the KMeans object does not have an attribute called `centroids_`. The correct attribute is `cluster_centers_`.

`print(kmeans.cluster_centers)` - This option is incorrect because it is missing the parentheses `_` after `cluster_centers`, which would result in a syntax error.

`print(kmeans.centroids)` - This option is incorrect because there is no attribute called `centroids` in the KMeans object. The correct attribute is `cluster_centers_`.

Question 7/20

To build an accurate model, there are methods we can use to determine the optimal number of clusters to employ in our analysis.

Which of the following measures are NOT typically used to help us determine the best K value for our data?

☐ Within-cluster variation

☐ The CH index

☐ Between-cluster variation

☒ Outer-cluster variation

Explanation:

Within-cluster variation: This option is incorrect because within-cluster variation (also known as within-cluster sum of squares or WCSS) is a commonly used measure to assess the compactness of clusters and to find the optimal number of clusters in k-means clustering.

Between-cluster variation: This option is incorrect

because between-cluster variation (also known as between-cluster sum of squares) is a crucial measure that evaluates the separation or distinctiveness of clusters and aids in determining the optimal number of clusters.

The CH index: This option is incorrect because the CH index (Calinski-Harabasz index) evaluates the quality of clustering based on both the within-cluster and between-cluster variation. It is also used to determine the optimal number of clusters, so it does help in finding the best K value for our data.

Outer-cluster variation: This option is correct because "outer-cluster variation" is NOT a standard measure used in cluster analysis and is not recognised in the context of cluster analysis or K-means clustering.

Question 8/20

Let's determine the optimal k value. Use the Elbow method, with WCSS, to find the best K value for our dataset.

Based on the elbow method plot, which is the best K value for our data?

☐ 3 clusters

☒ 5 clusters

☐ 7 clusters

☐ 10 clusters

Explanation:

3 clusters: This option is incorrect because it suggests choosing 3 clusters, which does not align with the elbow point in the plot's observation. The elbow point typically indicates the optimal number of clusters, and in ; in this case, it appears at 5 clusters.

7 clusters: This option is incorrect because it suggests choosing 7 clusters, which is not supported by the elbow method plot. The plot indicates an apparent kink at 5 clusters, implying that 7 clusters may not be the best choice.

10 clusters: This option is incorrect because it suggests choosing 10 clusters, which the elbow method plot does not support. The plot indicates an apparent kink at 5 clusters, implying that 10 clusters may result in overfitting and suboptimal clustering.

Question 9/20

Let's evaluate the model's quality by computing its silhouette score.

Based on these results, what is the silhouette score for a k-means model with $k=5$ clusters?

☐ 0.625

☐ 0.890

☒ 0.559

☐ 0.715

Explanation:

0.625: This option is incorrect because it suggests a silhouette score of 0.625, which is higher than expected and may indicate better clustering than the actual result.

0.715: This option is incorrect because it proposes a silhouette score of 0.715, which is higher than the correct answer. While a higher silhouette score is desirable, a value of 0.715 would imply even better clustering performance than what the model's results indicate.

0.890: This option is incorrect because it proposes a silhouette score of 0.890, higher than the correct answer. While a higher silhouette score is preferable, a value of 0.890 would suggest exceptionally good clustering performance, which does not align with the actual result.

Question 10/20

What does the resulting silhouette score indicate?

☐

The silhouette score does not really matter.

☐

It is quite bad because it is less than 10.

☐

It is very bad because we want it to be closer to -1.



It is good because it is greater than 0 and relatively close to 1.

Explanation:

"It is quite bad because it is less than 10." This is incorrect because a silhouette score ranges from -1 to 1. A score greater than 0 and closer to 1 indicates good clustering. The value "10" is irrelevant for silhouette scores.

"It is very bad because we want it to be closer to -1." This is incorrect because a silhouette score closer to -1 indicates poor clustering, where data points are likely assigned to the wrong clusters. A score closer to 1 is preferable.

"The silhouette score does not really matter." This is incorrect because the silhouette score is a widely used metric to evaluate the quality of clustering. It provides meaningful insights into how well clusters are formed.

Question 11/20

What is the next appropriate step after determining that the silhouette score for the optimal number of clusters, determined by the elbow method, is deemed unsatisfactory?

☐

Increase the number of clusters and recompute the silhouette score.

☐

Decrease the number of clusters and recompute the silhouette score.

☐

Ignore the silhouette score and proceed with the current clustering configuration.



Use the silhouette score as input into decision-making when evaluating other clustering algorithms.

Explanation:

"Increase the number of clusters and recompute the silhouette score." This option is incorrect because the elbow method was quite clear on the optimal number of clusters. Increasing the number of clusters may not necessarily improve the silhouette score. It could lead to overfitting and worsen clustering performance.

"Decrease the number of clusters and recompute the silhouette score." This option is incorrect because the elbow method was quite clear on the optimal number of clusters. Decreasing the number of clusters may not address the underlying issue causing the low silhouette score. It could oversimplify the clustering structure and disregard important patterns in the data.

"Ignore the silhouette score and proceed with the current clustering configuration." This option is incorrect because ignoring the silhouette score does not address the unsatisfactory clustering performance. It is important to assess and improve clustering quality to derive meaningful insights from the data.

Question 12/20

Hierarchical clustering could be a natural next step for further analysis, especially considering the presence of the outliers and the potentially uneven cluster shapes observed in the KMeans model results. Hierarchical clustering can offer insights into the hierarchical structure of the data, potentially capturing nuances that KMeans might overlook.

Which of the following is a key difference between KMeans and Hierarchical clustering algorithms?



Hierarchical Clustering forms spherical clusters around centroids. Each data point belongs to the cluster with the nearest centroid. In contrast, KMeans builds a hierarchy of clusters, starting with individual data points and gradually merging them into larger clusters based on proximity.



Hierarchical clustering does not require the number of clusters to be predefined. It produces a hierarchical tree of clusters, and the number of clusters can be determined after the clustering process, whereas KMeans requires the number of clusters (K) to be specified in advance.



Hierarchical Clustering requires the number of clusters (K) to be specified in advance. In contrast, KMeans does not require the number of clusters to be predefined. It produces a hierarchical tree of clusters, and the number of clusters can be determined after the clustering process.



Hierarchical clustering assumes clusters to be spherical and of equal size, which may not always reflect the true underlying structure of the data, whereas KMeans can capture clusters of arbitrary shapes and sizes, offering more flexibility in cluster formation.

Explanation:

"Hierarchical Clustering forms spherical clusters around centroids. Each data point belongs to the cluster with the nearest centroid, whereas KMeans builds a hierarchy of clusters, starting with individual data points and gradually merging them into larger clusters based on proximity." This option incorrectly describes the clustering processes of KMeans and Hierarchical Clustering. In fact, KMeans forms spherical clusters around centroids, while Hierarchical Clustering does not use centroids and does not necessarily result in spherical clusters.

"Hierarchical clustering assumes clusters to be spherical and of equal size, which may not always reflect the true underlying structure of the data, whereas KMeans can capture clusters of arbitrary shapes and sizes, offering more flexibility in cluster formation." This option inaccurately describes the assumptions made by Hierarchical Clustering and KMeans regarding cluster shapes and sizes. KMeans assumes clusters to be spherical and of equal size, while Hierarchical Clustering can accommodate clusters of arbitrary shapes and sizes.

"Hierarchical Clustering requires the number of

clusters (K) to be specified in advance, whereas KMeans does not require the number of clusters to be predefined. It produces a hierarchical tree of clusters, and the number of clusters can be determined after the clustering process." This option incorrectly describes the requirements for specifying the number of clusters in KMeans and Hierarchical Clustering. In reality, Hierarchical Clustering does not require the number of clusters to be predefined, while KMeans does require the number of clusters (K) to be specified in advance.

Question 13/20

What is the most appropriate number of clusters for the data points represented by the following dendrogram?

☐ 8

☐ 6

☒ 4

☐ 2

Explanation:

Observing the dendrogram can help determine the number of clusters that best depict different groups. The best number of clusters is the number of vertical lines in the dendrogram cut by a horizontal line that can transverse the maximum distance vertically without intersecting a cluster. In this example, the best number of clusters is 4.

2 is incorrect because the dendrogram suggests that the optimal number of clusters is not 2 but 4.

6 is incorrect because the dendrogram suggests that the optimal number of clusters is not 6 but 4.

8 is incorrect because the dendrogram suggests that the optimal number of clusters is not 8 but 4.

Question 14/20

The code attempts to perform agglomerative clustering on the scaled data, based on 5 clusters, using the AgglomerativeClustering class from sklearn and print the resulting cluster labels. However, there is an error in the code.

Which of the following options will fix the error and produce the desired results?

☐ Replace labels = fit_predict(X_scaled) with labels = cluster.fit(X_scaled)

☐ Replace labels = fit_predict(X_scaled) with labels = cluster.predict(X_scaled)

☒ Replace labels = fit_predict(X_scaled) with labels = cluster.fit_predict(X_scaled)

☐ Replace labels = fit_predict(X_scaled) with labels = cluster.fit_transform(X_scaled)

Explanation:

Replace labels = fit_predict(X_scaled) with labels = cluster.predict(X_scaled): This option is incorrect because AgglomerativeClustering does not have a predict method. Instead, it has a fit_predict method to fit the model to the data and return cluster labels for each sample.

Replace labels = fit_predict(X_scaled) with labels = cluster.fit(X_scaled): This option is incorrect because the fit method of AgglomerativeClustering does not return cluster labels. It only fits the model to the data without making predictions.

Replace labels = fit_predict(X_scaled) with labels = cluster.fit_transform(X_scaled): This option is incorrect because the fit_transform method of AgglomerativeClustering is used for dimensionality reduction, not for clustering. It fits the model to the data and transforms the data, but it does not return cluster labels.

Question 15/20

After creating a scatter plot visualising the clusters based on the cluster labels in question 14, we calculated and printed the mean values of each feature within each cluster, providing insights into the characteristics of each cluster. See image in the notebook.

Which of the following best describes the characteristics of Cluster 3 based on the mean values of each feature?

☐ Cluster 3 represents customers with average annual income and spending scores.

☒ Cluster 3 comprises data points with above-average annual income and spending scores.

☐ Cluster 3 consists of data points with below-average annual income and spending scores.

☒ Cluster 3 includes customers with significantly low annual income and spending scores.

Explanation:

Explanations: "Cluster 3 consists of data points with below-average annual income and spending scores." Incorrect because Cluster 3 has above-average values for both annual income and spending scores.

"Cluster 3 represents customers with average annual income and spending scores." Incorrect because Cluster 3's characteristics are described as above-average, not average as the values are significantly above the rest of the clusters.

"Cluster 3 includes customers with significantly low annual income and spending scores." Incorrect because the interpretation contradicts the resulting cluster profiles, which indicate above-average values for both annual income and spending scores in Cluster 3.

Question 16/20

Which of the following practical interpretations best describes Cluster 2 based on the mean values of each feature generated in question 15?

Customers in Cluster 2, on average, earn



more annually compared to the average earnings of all customers in the dataset.

☐

Customers in Cluster 2, on average, have lower annual income compared to the average earnings of all customers in the dataset.

☐

Customers in Cluster 2, on average, spend more compared to the average spending habits of all customers in the dataset.

☐

Customers in Cluster 2, on average, spend more compared to all the customers in the dataset because they earn more annually.

Explanation:

"Customers in Cluster 2, on average, have lower annual income compared to the average earnings of all customers in the dataset." This option is incorrect because it suggests that customers in Cluster 2 have lower annual income compared to the rest of the data, which contradicts the actual results showing a scaled average annual income of 0.609 which is higher than the lowest average of 0.33.

"Customers in Cluster 2, on average, spend more compared to the average spending habits of all customers in the dataset." This option is incorrect because it suggests that customers in Cluster 2 have a higher spending score compared to the rest of the data, which contradicts the actual results showing a scaled average spending score of 0.149 which is the lowest out of all the clusters.

"Customers in Cluster 2, on average, spend more compared to all the customers in the dataset because they earn more annually." This option is incorrect because it assumes that customers in Cluster 2 spend more solely based on their higher annual income, however, based on the results, cluster 2 customers have the lowest spending score.

Question 17/20

In order to understand the distribution of data points among different clusters we can visualise the number of data points per cluster using a bar chart.

Generate a bar chart that displays the number of data points in each cluster for both our features.

What inference can be drawn from the distribution of data points among clusters in our resulting clustering model?

☐

The clustering model demonstrates perfect separation of clusters, indicating robustness and accuracy.



The uneven distribution suggests potential issues with the clustering model's performance, such as overfitting.

☐

The imbalanced distribution suggests that the clustering algorithm requires more data to achieve better performance.

☐

The clustering model has successfully separated the data into distinct and balanced clusters.

Explanation:

The clustering model has successfully separated the data into distinct and balanced clusters: The statement is misleading as it implies that the clustering model has achieved perfect balance and separation, which may not be the case given the uneven distribution of data points among clusters.

The clustering model demonstrates perfect separation of clusters, indicating robustness and accuracy: This option overstates the performance of the clustering model by suggesting perfect separation, which is unlikely based solely on the distribution of data points among clusters.

The imbalanced distribution indicates that the clustering algorithm requires more data to achieve better performance: The imbalanced distribution of data points does not necessarily indicate a lack of data but rather suggests potential issues with the clustering model's ability to accurately capture the

underlying structure of the dataset.

Question 18/20

When evaluating the quality of clustering using the Davies-Bouldin Index (DBI), which of the following interpretations is correct?

☐ The DBI value does not provide any meaningful information about the clustering quality.

☒ A lower DBI value indicates better clustering, where clusters are more compact and well-separated.

☐ A higher DBI value indicates better clustering, indicating more diverse and spread-out clusters.

☐ The DBI value represents the average distance between all pairs of points in the dataset, with higher values indicating better clustering performance.

Explanation:

"A higher DBI value indicates better clustering, indicating more diverse and spread-out clusters." A higher DBI value actually suggests poorer clustering, as it indicates higher inter-cluster distances relative to intra-cluster distances, which implies less well-separated clusters.

"The DBI value does not provide any meaningful information about the clustering quality." This is incorrect. The DBI is specifically designed to measure the quality of clustering by considering both intra-cluster compactness and inter-cluster separation.

"The DBI value is used to measure the density within clusters, with higher values indicating denser clusters." The DBI is not a measure of average distance between all pairs of points in the dataset. It specifically evaluates the quality of clustering based on intra-cluster compactness and inter-cluster separation.

Question 19/20

Which linkage criterion measures the shortest distance between any two points in the two clusters being merged in hierarchical clustering?

☐ Complete linkage

☒ Single linkage

☐ Ward linkage

☐ Average linkage

Explanation:

Complete linkage is incorrect because it measures the maximum distance between any two points in the two clusters being merged.

Average linkage is incorrect because it measures the average distance between any two points in the two clusters being merged.

Ward linkage is incorrect because it measures the increase in variance when two clusters are merged.

Question 20/20

What is the best next step to take based on the silhouette scores for our KMeans clustering and Hierarchical clustering models, along with the distribution of data points in our clusters?

☐ Perform further exploratory data analysis to understand the underlying patterns and potential reasons for the distribution of data points.

☐ Refit both clustering models with different hyperparameters.

☐ Including additional relevant features to



enrich the dataset and improve the clustering models' performance.



All of the above

Explanation:

Refitting both clustering models with different hyperparameters is correct because by tweaking the parameters of both KMeans and Hierarchical clustering algorithms, it's possible to improve their performance and achieve stronger clustering quality.

Performing further exploratory data analysis to understand the underlying patterns and potential reasons for the distribution of data points is correct because this deeper analysis can provide insights into why certain clusters may have more or fewer data points and help identify any underlying trends or patterns that the clustering algorithms may have missed.

Including additional relevant features to enrich the dataset and improve the clustering models' performance is correct because by expanding the feature set, it's possible to capture more nuances and complexities in the data, which can lead to better clustering results. Adding relevant features may provide the algorithms with more information to distinguish between different clusters more effectively, ultimately enhancing the performance of the clustering models.